

Chapter 9:

Infinite Series

Section 9.7: Taylor Polynomials and Approximations

Taylor and Maclaurin Polynomials

REMARK Maclaurin polynomials are special types of Taylor polynomials for which $c = 0$.

Definitions of *n*th Taylor Polynomial and *n*th Maclaurin Polynomial

If f has n derivatives at c, then the polynomial

$$P_n(x) = f(c) + f'(c)(x - c) + \frac{f''(c)}{2!}(x - c)^2 + \cdots + \frac{f^{(n)}(c)}{n!}(x - c)^n$$

is called the **nth Taylor polynomial for f at c.** If $c = 0$, then

$$P_n(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots + \frac{f^{(n)}(0)}{n!}x^n$$

is also called the **nth Maclaurin polynomial for f.**

Remark: The 1st Taylor polynomial $P_1(x)$ of the function f at c is just the linearization of f that you have learned in MATH 101 section 4.8.

EXAMPLE

Find the Taylor polynomials P_0, P_1, P_2, P_3 and P_4 for $f(x) = \ln x$ centered at $c = 1$.

SOLUTION

Therefore, the Taylor polynomials are as follows.

$$P_0(x) = f(1) = 0$$

$$P_1(x) = f(1) + f'(1)(x - 1) = (x - 1)$$

$$P_2(x) = f(1) + f'(1)(x - 1) + \frac{f''(1)}{2!}(x - 1)^2$$

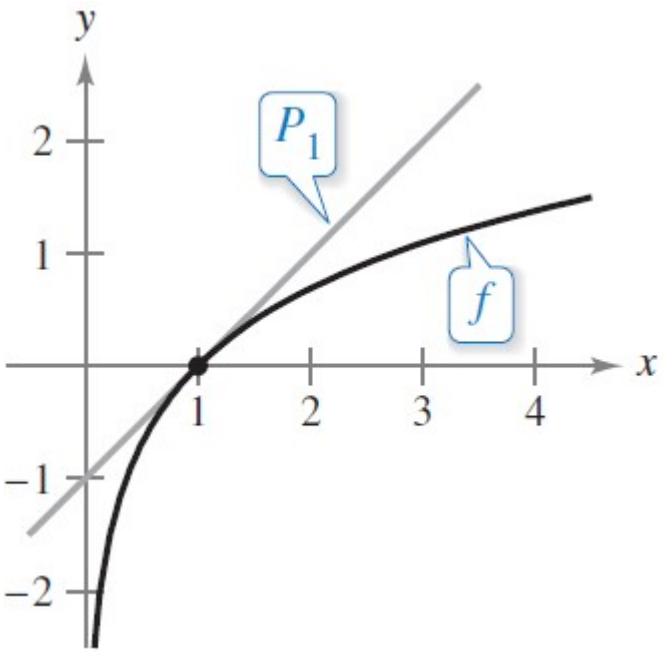
$$= (x - 1) - \frac{1}{2}(x - 1)^2$$

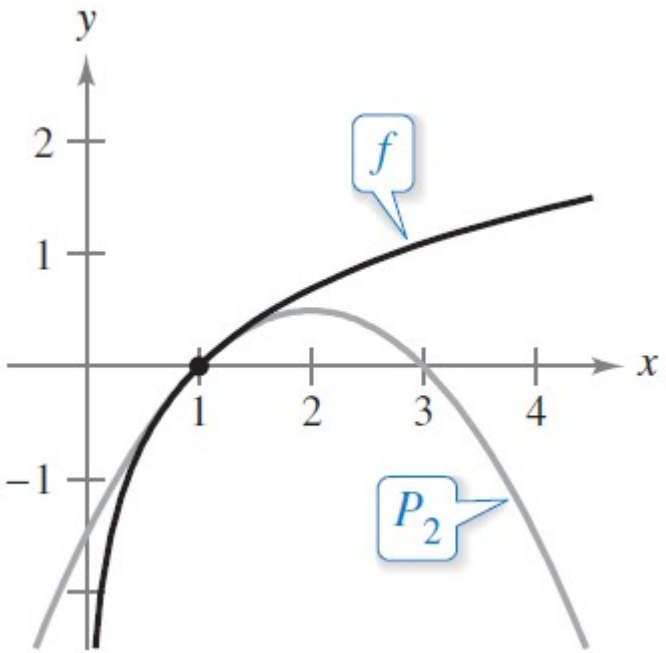
$$P_3(x) = f(1) + f'(1)(x - 1) + \frac{f''(1)}{2!}(x - 1)^2 + \frac{f'''(1)}{3!}(x - 1)^3$$

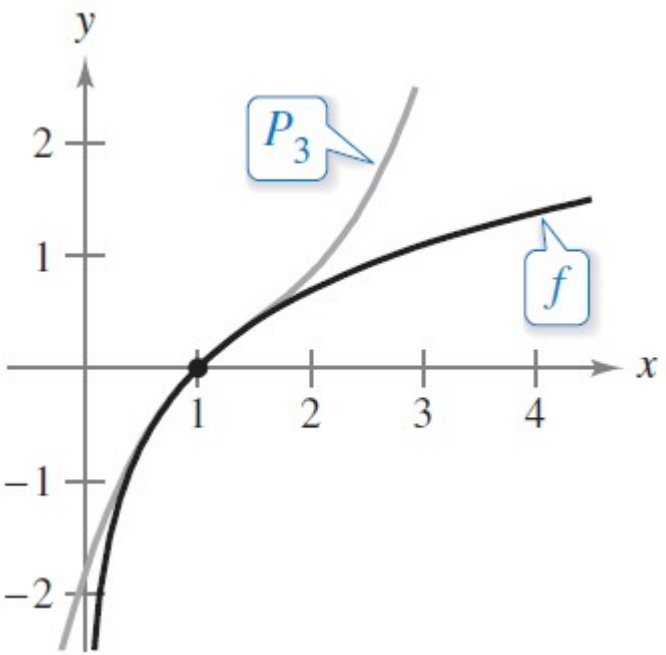
$$= (x - 1) - \frac{1}{2}(x - 1)^2 + \frac{1}{3}(x - 1)^3$$

$$P_4(x) = f(1) + f'(1)(x - 1) + \frac{f''(1)}{2!}(x - 1)^2 + \frac{f'''(1)}{3!}(x - 1)^3 + \frac{f^{(4)}(1)}{4!}(x - 1)^4$$

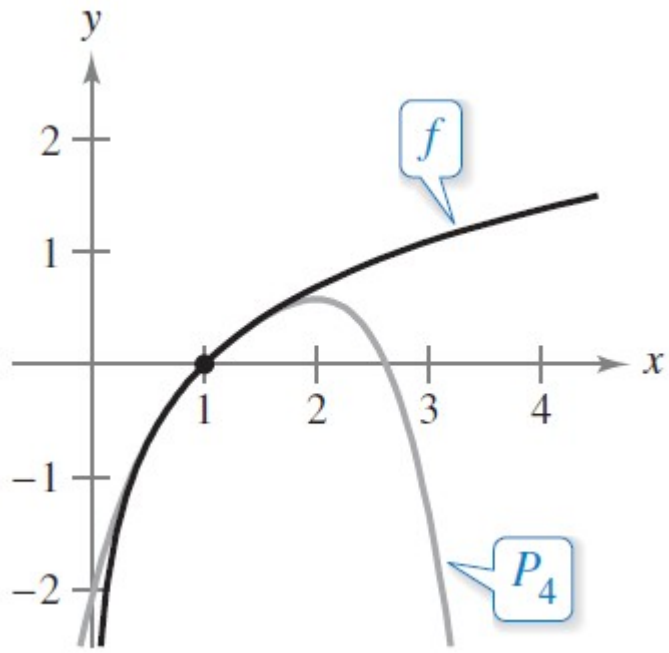
$$= (x - 1) - \frac{1}{2}(x - 1)^2 + \frac{1}{3}(x - 1)^3 - \frac{1}{4}(x - 1)^4$$







As n increases, the graph of P_n becomes a better and better approximation of the graph of $f(x) = \ln x$ near $x = 1$.



EXAMPLE

Find the Maclaurin polynomial P_6 for $f(x) = \cos x$ and use it to approximate the value of $\cos(0.1)$.

SOLUTION

$$P_6(x) = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6.$$

$$P_6(0.1) \approx 0.995004165$$

EXAMPLE

Finding a First-Degree Polynomial Approximation In Exercises 9–12, find a first-degree polynomial function P_1 whose value and slope agree with the value and slope of f at $x = c$. Use a graphing utility to graph f and P_1.

10. $f(x) = \dfrac{6}{\sqrt[3]{x}}, \quad c = 8$

SOLUTION

$$P_1(x) = -\frac{1}{8}x + 4$$

Solve exercises of section 9.7.

Start with the suggested problems mentioned in the syllabus.